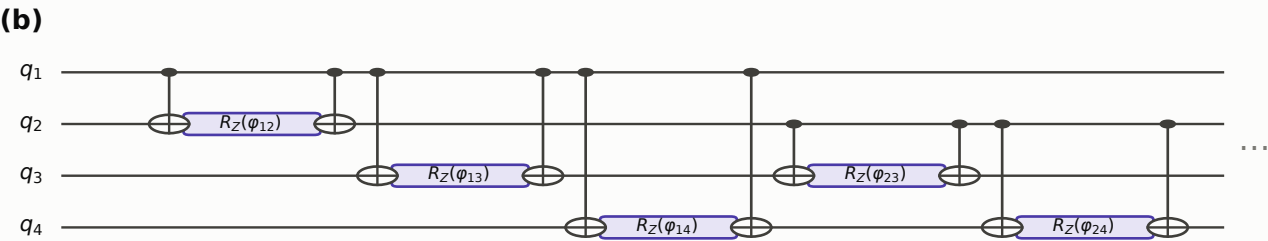
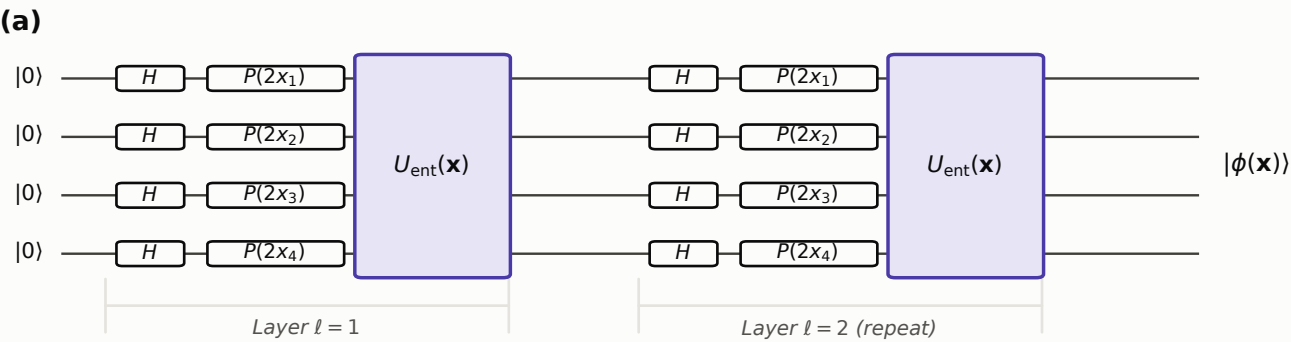


ZZFeatureMap circuit, $n = 4$ qubits, $r = 2$ repetitions, full entanglement



Pair (3, 4) omitted for space; the pattern is identical, $\text{CNOT}_{34} R_Z(\varphi_{34}) \text{CNOT}_{34}$.
Full entanglement on 4 qubits gives $\binom{4}{2} = 6$ trios per layer, so 24 CNOTs in total at $r = 2$.
 $\varphi_{ij}(\mathbf{x}) = 2(\pi - x_i)(\pi - x_j)$ encodes the pairwise feature interaction into the quantum phase.